STAT GU4204 — HW3 Solutions

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Due: 04/28/2026

All problems from DeGroot & Schervish, Probability and Statistics*, 4th ed.*

Problem 8.5.2(a)

Setting. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with both μ and σ^2 unknown. Construct a confidence interval for μ with confidence coefficient γ, using the sample mean

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad (S'_n)^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X}_n)^2.$$

Step 1: Identify a pivot. From the Fisher–Cochran theorem (proved in §8.3–8.4), $\bar{X}_n$ and $(S'_n)^2$ are independent, with

$$\sqrt{n}\,\frac{\bar{X}_n - \mu}{\sigma} \sim N(0,1), \qquad \frac{(n-1)(S'_n)^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Dividing the $N(0,1)$ random variable by $\sqrt{(n-1)(S'_n)^2/\sigma^2/(n-1)} = S'_n/\sigma$ yields the t-pivot

$$T \;=\; \sqrt{n}\,\frac{\bar{X}_n - \mu}{S'_n} \;\sim\; t_{n-1}, \tag{1}$$

which is free of μ and σ^2 (comment: this is exactly what makes it a pivot—its distribution does not depend on the unknown parameters).

Step 2: Invert the pivot. Let $c = T_{n-1,(1+\gamma)/2}$ denote the $(1+\gamma)/2$ quantile of t_{n-1}. By symmetry of the t distribution,

$$\Pr(-c \le T \le c) = \gamma.$$

Rearranging inside the probability:

$$\begin{aligned} -c \le \sqrt{n}\,\frac{\bar{X}_n - \mu}{S'_n} \le c \;&\iff\; -c\,\frac{S'_n}{\sqrt{n}} \le \bar{X}_n - \mu \le c\,\frac{S'_n}{\sqrt{n}} \\ &\iff\; \bar{X}_n - c\,\frac{S'_n}{\sqrt{n}} \le \mu \le \bar{X}_n + c\,\frac{S'_n}{\sqrt{n}}. \end{aligned}$$

Step 3: Confidence interval. Therefore the confidence interval with coefficient γ is

$$\boxed{\left(\bar{X}_n - T_{n-1,(1+\gamma)/2}\,\frac{S'_n}{\sqrt{n}},\; \bar{X}_n + T_{n-1,(1+\gamma)/2}\,\frac{S'_n}{\sqrt{n}}\right).} \tag{2}$$

Since the density of t_{n-1} is symmetric and unimodal about 0, the equal-tail choice is the *shortest* interval of the form $[-c, c]$ with coverage γ; hence (2) is the shortest t-based confidence interval for μ.

Remark. If the observed data yield $\bar{X}_n = \bar{x}$ and $S'_n = s'$, one simply plugs in: $\bar{x} \pm T_{n-1,(1+\gamma)/2}\, s'/\sqrt{n}$. For example, with $n = 9$, $\bar{x} = 22$, $s' = 3$, and $\gamma = 0.95$: $T_{8,0.975} \approx 2.306$, so the CI is $22 \pm 2.306 \cdot 3/3 = (19.694,\ 24.306)$.

Problem 8.5.4

Setting. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with both parameters unknown. Construct a confidence interval for σ^2 with confidence coefficient γ.

Step 1: Pivot. Recall $(n-1)(S'_n)^2/\sigma^2 \sim \chi^2_{n-1}$. Set

$$V \;=\; \frac{(n-1)(S'_n)^2}{\sigma^2} \;\sim\; \chi^2_{n-1}.$$

This is a pivot since its distribution does not depend on (μ, σ^2).

Step 2: Equal-tail cutoffs. Let $\chi^2_{n-1,\alpha}$ denote the α-quantile of χ^2_{n-1}, and set $\alpha = 1-\gamma$. Choose

$$a = \chi^2_{n-1,\alpha/2}, \qquad b = \chi^2_{n-1,1-\alpha/2}.$$

Then $\Pr(a \le V \le b) = \gamma$. (Comment: because the χ^2 density is not symmetric, equal-tail cutoffs do not minimize the CI length in general, but they are the standard textbook choice and yield a valid CI of coverage γ.)

Step 3: Invert for σ^2.

$$\begin{aligned} a \le \frac{(n-1)(S'_n)^2}{\sigma^2} \le b \;&\iff\; \frac{1}{b} \le \frac{\sigma^2}{(n-1)(S'_n)^2} \le \frac{1}{a} \\ &\iff\; \frac{(n-1)(S'_n)^2}{b} \le \sigma^2 \le \frac{(n-1)(S'_n)^2}{a}. \end{aligned}$$

Confidence interval.

$$\boxed{\left(\frac{(n-1)(S'_n)^2}{\chi^2_{n-1,\,1-\alpha/2}},\ \frac{(n-1)(S'_n)^2}{\chi^2_{n-1,\,\alpha/2}}\right), \quad \alpha = 1-\gamma.} \tag{3}$$

Taking square roots yields a confidence interval for σ:

$$\left(\sqrt{\frac{(n-1)(S'_n)^2}{\chi^2_{n-1,1-\alpha/2}}},\ \sqrt{\frac{(n-1)(S'_n)^2}{\chi^2_{n-1,\alpha/2}}}\right).$$

Problem 8.7.11(a)

Setting. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with both μ and σ^2 unknown, $n \geq 2$. Consider the class of estimators of σ^2 indexed by $c > 0$:

$$T_c \;=\; c \sum_{i=1}^{n} (X_i - \bar{X}_n)^2.$$

Part (a). Determine the value of c for which T_c is an unbiased estimator of σ^2.

Solution. From the distribution theory of the sample variance,

$$\frac{1}{\sigma^2} \sum_{i=1}^{n} (X_i - \bar{X}_n)^2 \;\sim\; \chi^2_{n-1}.$$

A χ^2_{n-1} random variable has mean $n-1$, so

$$\mathbb{E}\left[\sum_{i=1}^{n} (X_i - \bar{X}_n)^2\right] \;=\; (n-1)\sigma^2. \tag{4}$$

Therefore

$$\mathbb{E}[T_c] \;=\; c \cdot (n-1)\sigma^2.$$

For T_c to be unbiased for σ^2 we need $\mathbb{E}[T_c] = \sigma^2$, i.e.

$$c(n-1) = 1 \iff \boxed{c = \frac{1}{n-1}\,.}$$

This gives the familiar unbiased sample variance $(S'_n)^2 = \frac{1}{n-1} \sum (X_i - \bar{X}_n)^2$.

Remark (why not $c = 1/n$?). The maximum-likelihood estimator uses $c = 1/n$, which has smaller variance but nonzero bias $\mathbb{E}[T_{1/n}] - \sigma^2 = -\sigma^2/n$. The two estimators trade off bias and variance.

Problem 8.8.2

Setting. Compute the Fisher information $I(\theta)$ for a random sample of one observation from the exponential family

$$f(x \mid \theta) \;=\; \theta\, e^{-\theta x}, \qquad x > 0,\ \theta > 0.$$

(Equivalently: find the Fisher information about the rate parameter of an exponential random variable.)

Step 1: Log-likelihood and score.

$$\ell(x, \theta) \;=\; \log f(x \mid \theta) \;=\; \log \theta - \theta x.$$

The score function is

$$\dot{\ell}(x, \theta) \;=\; \frac{\partial \ell}{\partial \theta} \;=\; \frac{1}{\theta} - x.$$

Sanity check: $\mathbb{E}_\theta[\dot{\ell}(X, \theta)] = 1/\theta - \mathbb{E}_\theta[X] = 1/\theta - 1/\theta = 0$, as required for a regular model.

Step 2: Second derivative.

$$\ddot{\ell}(x,\theta) \;=\; \frac{\partial^2 \ell}{\partial \theta^2} \;=\; -\frac{1}{\theta^2}.$$

(Comment: $\ddot{\ell}$ does not depend on x, which simplifies the expectation.)

Step 3: Fisher information.

$$I(\theta) \;=\; -\mathbb{E}_\theta[\ddot{\ell}(X,\theta)] \;=\; -\left(-\frac{1}{\theta^2}\right) \;=\; \boxed{\frac{1}{\theta^2}.} \tag{5}$$

Cross-check via variance of score.

$$I(\theta) \;=\; \mathrm{Var}_\theta\left(\frac{1}{\theta} - X\right) \;=\; \mathrm{Var}_\theta(X) \;=\; \frac{1}{\theta^2}. \quad \checkmark$$

Information in a sample of size n. By additivity for iid samples, $I_n(\theta) = n\,I(\theta) = n/\theta^2$, and the Cramér–Rao lower bound for an unbiased estimator of θ is θ^2/n.

Problem 8.8.7

Setting. Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$ with μ *known* and σ^2 unknown. Consider the maximum likelihood estimator of σ^2:

$$\hat{\sigma}^2_{\text{MLE}} \;=\; \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2.$$

Determine whether $\hat{\sigma}^2_{\text{MLE}}$ is an efficient estimator of σ^2.

Step 1: Unbiasedness. Since μ is known, $(X_i - \mu)/\sigma \sim N(0,1)$ iid, so

$$\frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \mu)^2 \;\sim\; \chi^2_n, \qquad \mathbb{E}\left[\sum_{i=1}^{n}(X_i - \mu)^2\right] = n\sigma^2.$$

Thus $\mathbb{E}[\hat{\sigma}^2_{\text{MLE}}] = \sigma^2$; the estimator is unbiased.

Step 2: Variance of $\hat{\sigma}^2_{\text{MLE}}$. Using $\mathrm{Var}(\chi^2_n) = 2n$,

$$\mathrm{Var}(\hat{\sigma}^2_{\text{MLE}}) \;=\; \frac{1}{n^2}\cdot\sigma^4\cdot\mathrm{Var}(\chi^2_n) \;=\; \frac{1}{n^2}\cdot\sigma^4\cdot 2n \;=\; \frac{2\sigma^4}{n}. \tag{6}$$

Step 3: Fisher information for σ^2. Let $\theta = \sigma^2$. For a single $X \sim N(\mu, \theta)$,

$$\ell(x,\theta) = -\tfrac{1}{2}\log(2\pi) - \tfrac{1}{2}\log\theta - \frac{(x-\mu)^2}{2\theta},$$

$$\dot{\ell}(x,\theta) = -\frac{1}{2\theta} + \frac{(x-\mu)^2}{2\theta^2}, \qquad \ddot{\ell}(x,\theta) = \frac{1}{2\theta^2} - \frac{(x-\mu)^2}{\theta^3}.$$

Taking $-\mathbb{E}[\cdot]$ with $\mathbb{E}[(X-\mu)^2] = \theta$:

$$I(\theta) \;=\; -\frac{1}{2\theta^2} + \frac{\theta}{\theta^3} \;=\; \frac{1}{2\theta^2} \;=\; \frac{1}{2\sigma^4}. \tag{7}$$

Step 4: Cramér–Rao lower bound. For an unbiased estimator of $\theta = \sigma^2$ (here $g(\theta) = \theta$, $g' = 1$),

$$\text{CRLB} \;=\; \frac{[g'(\theta)]^2}{n\,I(\theta)} \;=\; \frac{1}{n \cdot \frac{1}{2\sigma^4}} \;=\; \frac{2\sigma^4}{n}.$$

Step 5: Compare. From (6) and the CRLB,

$$\mathrm{Var}(\hat{\sigma}^2_{\text{MLE}}) \;=\; \frac{2\sigma^4}{n} \;=\; \text{CRLB}.$$

Since $\hat{\sigma}^2_{\text{MLE}}$ is unbiased and attains the Cramér–Rao lower bound,

$$\boxed{\hat{\sigma}^2_{\text{MLE}} = \tfrac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2 \text{ is an efficient estimator of } \sigma^2.}$$

Problem 8.8.17

Setting. Suppose $X_1, \ldots, X_n$ form a random sample from a distribution with density $f(x \mid \theta)$ satisfying the usual regularity conditions. Let $T = r(X_1, \ldots, X_n)$ be an *unbiased* estimator of $g(\theta)$. Show that T attains the Cramér–Rao lower bound

$$\mathrm{Var}_\theta(T) \;=\; \frac{[g'(\theta)]^2}{n\,I(\theta)}$$

if and only if there is a function $u(\theta)$ such that

$$T - g(\theta) \;=\; u(\theta)\sum_{i=1}^{n}\dot{\ell}(X_i, \theta) \qquad \text{a.s. } \Pr_\theta, \tag{8}$$

i.e. the centered estimator is proportional (in θ) to the total score.

Setup. Let $U_n(\theta) = \sum_{i=1}^{n}\dot{\ell}(X_i, \theta)$ denote the total score. Under regularity:

- $\mathbb{E}_\theta[U_n(\theta)] = 0$;
- $\mathrm{Var}_\theta(U_n(\theta)) = n\,I(\theta)$;
- For any unbiased estimator T of $g(\theta)$, differentiating $\mathbb{E}_\theta[T] = g(\theta)$ under the integral sign gives
$$\mathrm{Cov}_\theta(T, U_n(\theta)) \;=\; g'(\theta). \tag{9}$$

Cauchy–Schwarz / correlation form of CRLB. By Cauchy–Schwarz,

$$[g'(\theta)]^2 \;=\; [\mathrm{Cov}_\theta(T, U_n)]^2 \;\le\; \mathrm{Var}_\theta(T)\cdot\mathrm{Var}_\theta(U_n) \;=\; \mathrm{Var}_\theta(T)\cdot n\,I(\theta),$$

which is exactly the CRLB inequality.

Equality condition. Equality in Cauchy–Schwarz for random variables $Y = T - g(\theta)$ and $Z = U_n(\theta)$ (both with mean zero under θ) holds *iff* Y and Z are linearly dependent $\Pr_\theta$-almost surely: there exists $u(\theta) \in \mathbb{R}$ such that

$$T - g(\theta) \;=\; u(\theta)\,U_n(\theta) \qquad \text{a.s. } \Pr_\theta. \tag{10}$$

This is precisely (8).

Determining $u(\theta)$. Taking covariance of both sides of (10) with $U_n(\theta)$:

$$g'(\theta) \;=\; \mathrm{Cov}_\theta(T, U_n) \;=\; u(\theta)\,\mathrm{Var}_\theta(U_n) \;=\; u(\theta)\,n\,I(\theta),$$

so

$$u(\theta) \;=\; \frac{g'(\theta)}{n\,I(\theta)}.$$

In this case $\mathrm{Var}_\theta(T) = u(\theta)^2 \cdot nI(\theta) = [g'(\theta)]^2/(nI(\theta))$, confirming attainment of the CRLB.

Conclusion.

$$\boxed{\mathrm{Var}_\theta(T) = \frac{[g'(\theta)]^2}{n\,I(\theta)} \iff T - g(\theta) = \frac{g'(\theta)}{n\,I(\theta)} \sum_{i=1}^{n} \dot{\ell}(X_i, \theta) \;\text{ a.s.}}$$

Interpretation. Attaining the CRLB forces the centered estimator to be a (deterministic-in-θ) multiple of the score. In particular, T must be a function of the score, and hence of the sufficient statistic—an attainment-forces-sufficiency result.

Problem 9.1.1(a)

Setting. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, 1)$, and consider the simple-vs-simple testing problem

$$H_0 : \mu = 0 \qquad \text{vs.} \qquad H_1 : \mu = 1.$$

A test δ is defined by rejecting H_0 if and only if $\bar{X}_n > c$, where c is a constant. Part (a) asks: compute the probabilities of type I error (α) and type II error (β) as functions of c and n.

Sampling distributions. Under either hypothesis, $\bar{X}_n \sim N(\mu, 1/n)$, so

$$\sqrt{n}\,(\bar{X}_n - \mu) \;\sim\; N(0, 1).$$

Type I error probability (α, size of the test).

$$\begin{aligned}
\alpha(c) = \Pr_{\mu=0}(\text{Reject } H_0) \;&=\; \Pr_{\mu=0}(\bar{X}_n > c) \\
&= \Pr\big(\sqrt{n}\,(\bar{X}_n - 0) > \sqrt{n}\,c\big) \\
&= 1 - \Phi(\sqrt{n}\,c).
\end{aligned}$$

Type II error probability (β).

$$\begin{aligned}
\beta(c) = \Pr_{\mu=1}(\text{Accept } H_0) \;&=\; \Pr_{\mu=1}(\bar{X}_n \le c) \\
&= \Pr\big(\sqrt{n}\,(\bar{X}_n - 1) \le \sqrt{n}\,(c-1)\big) \\
&= \Phi\big(\sqrt{n}\,(c-1)\big)\,.
\end{aligned}$$

Summary.

$$\boxed{\alpha(c) \;=\; 1 - \Phi(\sqrt{n}\,c), \qquad \beta(c) \;=\; \Phi\big(\sqrt{n}\,(c-1)\big)\,.}$$

Comments.

- As $c \uparrow$, $\alpha \downarrow$ and $\beta \uparrow$—the usual error tradeoff.
- For a test of size α, choose $c = z_{1-\alpha}/\sqrt{n}$, where $z_{1-\alpha} = \Phi^{-1}(1-\alpha)$; then $\beta = \Phi\big(z_{1-\alpha}/\sqrt{n} \cdot \sqrt{n} - \sqrt{n}\big) = \Phi\big(z_{1-\alpha} - \sqrt{n}\big)$, which $\to 0$ as $n \to \infty$ (consistency).
- For instance, $n = 25$ and $c = 0.5$ give $\alpha = 1 - \Phi(2.5) \approx 0.0062$ and $\beta = \Phi(-2.5) \approx 0.0062$; the test is nearly symmetric in errors.